

Today lecture: Hammack and Jay Cumming

last class:

Theorem: a mathematical statement that is true and can be proven true,

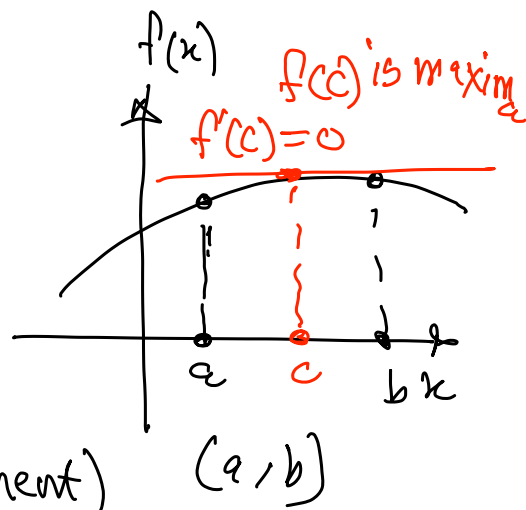
example:

(i) Theorem: if $\sum_{k=0}^{\infty} a_k$ converges then $\lim_{k \rightarrow \infty} a_k = 0$

(ii) Theorem: let f be a differentiable function on an open interval I , and let $c \in I$

if, $f(c)$ is maxima or minima of f in I then

$$f'(c) = 0$$



If P then Q $P \Rightarrow Q$

(conditional statement)

Theorem: a sequence in $\mathbb{R}$ is a convergent sequence if and only if it is a Cauchy sequence,

P if and only if Q , $P \Leftrightarrow Q$

(they are equivalent statements)

• Proposition:

A statement that is true but not as significant
the outcome of a proposition is not as

useful as a theorem.

• lemma:

A theorem whose main purpose is to prove other theorems

(Zorn's lemma $\Leftrightarrow$ Axiom of choice)

• corollary:

A result which is an immediate consequence of a theorem or proposition

(special cases of a theorem, proposition)

• conjecture:

A statement that someone believes to be true however no one can prove or disprove it.

• Goldbach conjecture (strong form)

Say:

"Every perfect number is even"

Natural you will ask:

• What is a perfect number?

• What do you mean by even numbers?

• Can negative numbers be even number?

Before dealing with statement we need some preambles:

some standards need to be set.

(we need to define things)

hence we require definitions

- they provide us with precision.

→ A challenge to precisely write down the description of sth:

say "What do you mean by a sandwich?"

→ sth that is stuffed in between two breads

↓ egg, lettuce, meat

- Grilled cheese sandwich: ^{two bread} with cheese inside.

- We have to know about bread, egg, lettuce, meat beforehand.

Also another ambiguity: egg fried, boiled, poached, raw?

- to prevent misconception and ambiguity we need to define thing precisely,

Every perfect number is even,

Precise information:

(i) perfect number

(ii) even number: 2, 4, 6, 8, 10, ...

(a) is 0 an even number?

(b) what about (-10)?

we consider term even, since our definition of even number is:

Defⁿ:

A number n is even, if $n = 2a$ for $a \in \mathbb{Z}$,

- 0 is even, since, $0 = 2 \cdot a$ that implies $a = 0$ (integer)
- -10 is even, $-10 = 2 \cdot a \Rightarrow a = -5$ (integer)
- -7 is not integer, $-7 = 2 \cdot a \rightarrow a = -7/2$ (not an integer)

Defⁿ:

An integer n is odd, if $n = 2k+1$ for some $k \in \mathbb{Z}$,

Defⁿ:

two integers have the same parity if they are both even or both odd otherwise opposite parity.

Just like theorem, Definition also look like:

if P then Q

however: they are actually biconditional statement,

example:

• If An integer 6 is even, then, $6 = 2 \cdot a$ for $a = 3$
('integer')

• if, $6 = 2 \cdot a$ for $a = 3$ (an integer)
then 6 is an even number,

Defn:

suppose a and b are integers, a divides b ($a|b$)

if $b = ac$ for some integer c ,

$\begin{cases} a \text{ is a divisor of } b \\ b \text{ is a multiple of } a \end{cases}$

Defn:

$n \in \mathbb{N}$, is called prime if n has exactly two
positive divisors 1 and n

Defn:

• The greatest common divisor, (gcd) of two integers
 a and b denoted by $\gcd(a, b)$ is the largest
integer that divides both a and b ,

Defn:

least common multiple of non zero integers
 a and b denoted by $\text{lcm}(a, b)$ is the smallest

integer in $\mathbb{N}$ that is a multiple of both a and b

example:

$$\gcd(18, 24) = 6, \quad \gcd(3, -6) = 3$$

$$\gcd(0, 7) = 7$$

$$\gcd(0, 0) = \boxed{\infty} \rightarrow \text{is not in the set of integers}$$

→ There is an ambiguity in the Defn of $\gcd$

→ at least one of a and b is non zero

→ in maths we want to be precise however not overly precisely,

say: odd $\Rightarrow 2k+1$ type number k infer

we didn't $+$ mean?

we didn't integer mean?

• $<, =, >$, meaning?

we know term naturally

$=$: equal to mean two things that are equal
(circular argument)

$<$:

we will take some things as fact:

$=, <, >$,

$$a, b \in \mathbb{Z}, \quad a+b \in \mathbb{Z}, \quad a^v \in \mathbb{Z}, \quad a \cdot b \in \mathbb{Z}$$

$$a \cdot b + 6a \in \mathbb{Z}$$

Direct proof:

We will focus mostly on proposition,

Proposition: 'if $P \Rightarrow Q$

our goal: to show this conditional statement is true

x

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

However, we will discard
 $P = \text{false}$

We start by assuming
 (i) $P = \text{true}$

(ii) and show Q turns out to be
 true

A general scheme:

Proof: If P then Q

suppose P is true

< explanation of what P means >

- algebra
- logic
- Defn
- ⋮

< keep looking, that's what Q means >

Hence Q is true

example:

Proposition: if x is odd then x^n is odd,

Proof: suppose x is odd

from Defn, $x = 2k+1$ for $k \in \mathbb{Z}$

Thus, $x^n = (2k+1)^n$

$$\Rightarrow x^n = 2k^n + 4k + 1$$

$$\Rightarrow x^n = 2(2k^n + 2k) + 1$$

since $k \in \mathbb{Z}$, $2k^n + 2k \in \mathbb{Z}$, we assume $2k^n + 2k = c$
 $c \in \mathbb{Z}$

Thus, $x^n = 2c + 1$ for c some integer

therefore x^n is odd,

Proposition: The sum of two even integers
is even,

Proof: suppose a and b two even integers
by Defn, $a = 2k_1$, $b = 2k_2$ [$k_1, k_2 \in \mathbb{Z}$]

$$\text{Now, } a + b = 2k_1 + 2k_2$$

$$\Rightarrow a + b = 2(k_1 + k_2)$$

since, k_1 and k_2 integers, $k_1 + k_2$ is also integer,
 $\therefore k = k_1 + k_2$ for, k is an integer

$$\text{Thus, } a + b = 2k, \quad k \in \mathbb{Z}$$

Hence $a + b$ is an even integer

Proposition: x and y positive number if $x \leq y$
then $\sqrt{x} \leq \sqrt{y}$

Proof: suppose: $x \leq y$, for x, y positive number
(we don't define $\leq$, positive number meaning)

Now, from $x \leq y$, we subtract y from both side,

$$\Rightarrow x - y \leq 0$$

$$\Rightarrow \underline{(\sqrt{x})^2 - (\sqrt{y})^2 \leq 0}$$

for x positive
 $x = (\sqrt{x})^2$

$$\Rightarrow (\sqrt{x} - \sqrt{y}) \underline{(\sqrt{x} + \sqrt{y})} \leq 0$$

Dividing both side with the positive number
 $(\sqrt{x} + \sqrt{y})$

$$\Rightarrow \sqrt{x} - \sqrt{y} \leq 0$$

$$\Rightarrow \sqrt{x} \leq \sqrt{y} \quad [\text{adding } \sqrt{y} \text{ on both sides}]$$